curriculum vitæ

Gavin Lucsik

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Profile

I am a physicist studying geometry and topology in quantum systems. I am interested in geometric representations of quantum information and the structure and symmetry that they inherit.

Education

BS Physics, California State University, Long Beach — Expected May 2027
GPA In-Major: 4.00

BS Mathematics, California State University, Long Beach — Expected May 2027
GPA In-Major: 4.00
Minor in Geology

Cumulative GPA: 4.00

Physics Courses Taken: Quantum Mechanics I & II, Electrodynamics I & II, Computational Physics, Thermodynamics, Geophysics
Math Courses Taken: Abstract Algebra, Differential Geometry, Real Analysis, Number Theory, ODE's, Probability

Research Experience

University of California, Santa Barbara
Quantum Foundry Researcher

Advisors: Prof. John Harter, Jack Tregidga | *Department of Materials & Physics* — May 2026 - August 2026

- Mapped topological phase transitions of ferroelectric superconductors using Density Functional Theory (DFT).
- Studied $k \cdot p$ low energy effective theory to model topological behavior.
- Performed calculations using VASP, WANNIER90, WannierTools on an HPC cluster.

California State University, Long Beach
Undergraduate Researcher

Advisors: Prof. Michael Peterson, Prof. Andreas Bill | *Department of Physics* — December 2024 - Present

- Studied the geometric dependence of topological phase transitions in low dimensional lattice models.
- Derived high dimensional topological phase transition boundaries parameterized by lattice distortions.
- Analytically solved for topological invariants including Berry phase, Zak phase, Chern number

California State University, Long Beach
Undergraduate Researcher

Advisor: Prof. Benjamin Hagedorn | *Department of Engineering & Earth Sciences* — August 2023 - November 2024

- Developed an algorithm to characterize components of groundwater flow derived from non-perturbative measurements.
- Integrated multivariate optimization to refine model predictions.
- Applied signal processing techniques to handle gaps and noise in centennial datasets.
- Developed a bounding technique to physically ground hydrograph separation results.

Professional Experience

Tetra Tech September 2024 - May 2026
Engineering Intern

- Processed large datasets from field measurements, laboratory analyses, and GIS sources for environmental site assessments.
- Developed an automated data management system to organize and track laboratory instruments and field equipment using python.
- Created technical schematics and site maps using CAD and GIS software.

Port of Long Beach August 2023 - September 2024
Engineering Intern

- Collected and processed geospatial data to model seafloor topography.
- Constructed high-resolution maps and 3D models using LiDAR, sonar, and drone imaging data.
- Operated and maintained Multi-beam LiDAR system aboard survey vessel

Technical Skills

Python, R, LaTeX, Git, Quarto, Adobe/GIMP photoshop, Blender, MS excel, AutoCAD.
DFT Software: VASP, WANIIER90, WannierTools
Experience running simulations on high-performance computing (HPC) clusters.

Presentations

Conference Abstracts

1. Lucsik, G., Tregidga, J., Harter, J., "Topological Phase Transitions in Ferroelectric-like Superconductors," California NanoSystems Institute Summer Conference, 13 August, 2026, Goleta CA.
2. Lucsik, G., Hagedorn, B., “Refining Baseflow Quantifications Using Multi-Objective Optimization and Digital Filtering," American Geophysical Union Fall Conference, 12 December, 2024, Washington D.C.

Honors and Awards

- The G2 Software Systems, Georgia Griffiths Endowed Scholarship May, 2026
- Barbara Turner & Kenneth K. Warner Endowed Award in Mathematics May, 2026
- The Richard and Florence Scalettar Scholarship in Physics May, 2026
- Partnership for Research in Materials Science Scholarship in Physics January, 2026
- LA Basin Geological Society Scholarship in Geophysics March, 2026
- John M. Turner Scholarship in Physics June, 2025
- Keung Luke Physics Assistanship May, 2025
- METRIC 2 Fellowship December, 2024 - Present
- COAST Field Experiences Award August, 2024
- CSULB President’s Honor List May, 2023 - Present

Affiliations

Society of Physics Students (Secretary) April 2025 - Present
The American Geophysical Union (Member) May 2023 – Present

References

Michael Peterson, Ph.D.
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